

Pedro Uribe

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Engineering student building edge AI, embedded, and robotics systems, from 4-bit quantized models designed for custom silicon to LiDAR and YOLO perception on an autonomous boat. Hackathon winner, world-level robotics champion, and founder of an AI product that has generated about US\$1,500.

EDUCATION

Tecnológico de Monterrey — *Monterrey, Mexico*

5th semester, in progress

CETYS Preparatoria — *International Baccalaureate, GPA 9.5/10*

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript. Spoken: Spanish (native), English (fluent)

Edge AI and vision: TensorRT, ONNX, YOLO, OpenCV, scikit-learn, model quantization, NVIDIA Jetson

Embedded and hardware: STM32, ESP32, FPGA (Xilinx ZYNQ), RTOS, UART, CAN

Robotics: ROS 2, Gazebo, LiDAR, PID and feedforward control, WPILib

Tools: Linux, Docker, Git, PostgreSQL, Redis, FastAPI, Flask

EXPERIENCE

USV Perception Lead, *VantTec (Tecnológico de Monterrey)*

2025 – Present

- Lead perception for an autonomous surface vehicle at RoboBoat 2026; rebuilt the ROS 2 vision pipeline after its stereo camera failed, fusing a Velodyne VLP-16 LiDAR with a YOLO detector to publish 3D object positions.
- Wrote the LiDAR-to-bounding-box ranging node (median range with MAD outlier rejection): ~6.2 m against ~7 m ground truth in simulation, and markers measured out to ~18 m on recorded boat data.
- Trained the YOLO buoy dataset, compiled the TensorRT engine, and validated in Gazebo Harmonic before water tests; built SocketCAN, Xbee telemetry, and remote-terminal links for field testing.

Research Assistant, *University of California San Diego (VLSI Information Processing Lab)*

Jun – Aug 2023

- Designed a real-time shape classifier (linear SVM) with weights quantized to 4-bit integers for hardware implementation, holding 98%+ accuracy on 3,000 test images; targeted a Xilinx ZYNQ board.
- Presented at the UCSD closing conference and as an invited speaker at the UCSD-ENLACE-México Symposium (Instituto Tecnológico de Tijuana).

Software Lead and Captain, *Innobotics FRC Team 6694*

Feb 2022 – Jan 2025

- Owned robot code in Java/WPILib over three seasons: subsystems, autonomous routines, PID and feedforward control tuned from SysId measurements, and NetworkTables telemetry.
- Led the team to 3rd place and the Spirit Award at Regional Hermosillo (2024) and 5th place with the Imagery Award at Regional Monterrey (2023).

Software Lead and Captain, *Mexicali Robotics*

2016 – 2022

- Built a STEM and electronics foundation through six years of competition; 3rd place worldwide at Infomatrix World Finals, Bucharest (2019).

PROJECTS

SentinelBox, edge-AI HVAC predictive maintenance *1st place, Carrier Keep It Cool Hackathon 2026*

github.com/ayexxxd/SentinelBox

- Designed the electronics and UART link between sensor nodes and an ESP32 gateway that learns each unit's baseline and runs a compact neural network on-device, with no cloud dependency.
- Controlled prototype validation: 6 s average detection time and 0% false alarms after a 90 s baseline learning period.

Uribot, AI automation platform for small businesses *Founder*

2025 – Present | uribot.chat

- Built a multi-tenant system (Python, Flask, PostgreSQL, Redis, Docker) that turns WhatsApp, Instagram, and Messenger chats into quotes and bookings with human handoff; about US\$1,500 in revenue to date.

MiNorte, AI financial analyst for small businesses *HackMTY 2026, Banorte Challenge*

github.com/pedroolib/MiNorte-by-Banorte

- Built the deterministic financial engine, FastAPI backend, and MCP tools that let AI agents explain verified numbers and generate dashboard cards.

Smart Tractor, embedded vision

github.com/pedroolib/JohnDeere-RLTractorModelation

- STM32/RTOS tractor controlled with Logitech pedals, with a Grafana dashboard and vision-based operator drowsiness detection.

HONORS AND LEADERSHIP

- 1st place worldwide, NASEF Minecraft Masters (Dec 2020) and NASEF Farmcraft (Jul 2020), representing Mexico
- 3rd place worldwide, Lego Sumo, Infomatrix World Finals (2019); 2nd place continental, Robomatrix (2018)
- 3rd place regional and Spirit Award, FIRST Robotics Competition, Hermosillo (2024)
- FTC robotics mentor, Botbusters (2025); robotics education volunteer, SMALC (2026)